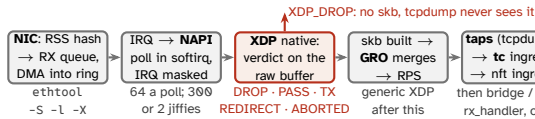
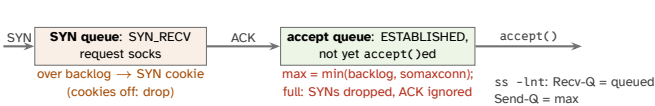


In one line: A frame is DMA'd into a **ring**, polled by **NAPI** in softirq, offered to **XDP**, merged by **GRO**, copied to **taps**, run through **tc** and the **netfilter** hooks; a **routing decision** sends it *up* to a socket (INPUT) or *across* (FORWARD). **contrack** remembers every flow, so NAT is decided on the **first** packet and replies are reversed for free. **DNAT before routing, SNAT after**.

RX: driver + device layer



A listening TCP socket has two queues



Receive: NAPI, offloads, softirq

- One IRQ schedules **NAPI**; the IRQ stays masked while the driver is polled. Caps: dev_weight, netdev_budget, netdev_budget_usecs (sysctl since 4.12).
- /proc/net/softnet_stat: one hex row per CPU — col 1 processed, col 2 **dropped** (backlog above netdev_max_backlog 10000), col 3 **time_squeeze** (budget ran out with work left).
- GRO** merges a flow's segments into one big skb; **TSO** (NIC) and **GSO** (software) cut it again on the way out — GRO is GSO's inverse. ethtool -k lists offloads.
- Threaded NAPI: echo 1 > /sys/class/net/<dev>/threaded. Busy polling: SO_BUSY_POLL, net.core.busy_poll; napi_defer_hard_irqs + gro_flush_timeout keep the IRQ off longer.

Spreading over CPUs

RSS	NIC hashes headers (usually Toeplitz) into an indirection table ($\geq 4 \times$ queues) → RX queue + IRQ	ethtool -X, IRQ affinity
RPS	software RSS: skb→hash picks the CPU whose backlog gets it; off until set	rx-<n>/rps_cpus
RFS	RPS aimed at the CPU where the reading thread runs	rps_sock_flow_entries (65536) + rps_flow_cnt
arFS	the NIC itself steers to that CPU	driver + ntuple filters
XPS	TX queue chosen per CPU or per RX queue; driver may set it	tx-<n>/xps_cpus, xps_rxqs

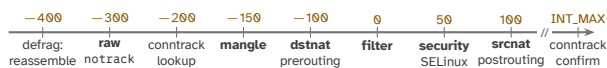
eBPF in the data path

- XDP** (4.8; “not kernel bypass” — PASS continues into the stack): native in the driver (ip link set dev eth0 xdpdrv obj p.o), xdpgeneric on an skb after allocation (any NIC, slower), xdpoffload on the NIC. TX = back out the same port; REDIRECT = another NIC, a CPU or an AF.XDP socket.
- AF_XDP** (4.18): REDIRECT through an XSKMAP into a socket whose **UMEM** userspace shares via FILL, COMPLETION, RX and TX rings; zero-copy if the driver supports it, else copy mode.
- tc**: clsact ingress/egress classifiers see an skb, return verdicts in direct-action mode; **tcx** (6.6) attaches them as BPF links; **netkit** (6.7) is a device with BPF in its transmit path, for containers.

nftables vs iptables

- iptables**: fixed tables with built-in chains — filter (INPUT FORWARD OUTPUT), nat (PRE INPUT OUTPUT POST), mangle (all five), raw (PRE OUTPUT), security (INPUT FORWARD OUTPUT); one tool per family; legacy -A fetches, edits and reloads the whole table.
- nftables**: nothing predefined. Tables per family (ip ip6 inet arp bridge netdev); base chains name type (filter anywhere; nat at pre/input/output/post; route at output), hook, priority, policy (default accept). Sets, maps, verdict maps, concatenations; nft -f swaps the ruleset in one atomic operation.
- Debian 10 (2019) and RHEL 8 ship iptables as **iptables-nft**: old syntax, nf_tables backend, iptables -V says (nf_tables); firewalld defaults to nftables since 0.6.0. Legacy rules loaded beside it are invisible to nft list ruleset — only a “iptables-legacy tables present” warning.

Hook priorities — lower runs first



Bridge family: dstnat — 300, filter — 200, out 100, srcnat 300. Every base chain on a hook runs, in priority order, whatever its table: **accept** ends only the current chain — the next chain and later hooks still judge; **drop** ends everything.

Sysctls

core.somaxconn 4096	caps Listen() backlog (128 before 5.4)
tcp_synccookies 1	cookies when the SYN queue overflows
tcp_max_syn_backlog	≥ 128 , by RAM; only used with cookies off
tcp_abort_on_overflow 0	1 = RST when the accept queue is full
tcp_tw_reuse 2	0 off · 1 outgoing · 2 loopback only
ip_forward 0	a write resets every IPv4 conf/* value
ipv6.conf.all.forwarding	1 ignores RAs unless accept_ra=2
ip_local_port_range	32768-60999

contrack and NAT

- One entry per flow, hashed twice (original + reply tuple). ct state: new, established (reply seen), related (ICMP error, FTP data), invalid, untracked. Rule 1 of a stateful ruleset: ct state established,related accept.
- NAT rules are looked up for the **first** packet only; the binding is stored in the entry and replies are reversed. **masquerade** = SNAT to the outgoing interface's current address (postrouting only); **redirect** = DNAT to this host. NAT in inet tables since 5.2; iptables and nft NAT coexist since 4.18.
- Size: buckets = RAM / 16384 (1024-262144), nf_contrack_max = buckets. Full → “nf_contrack: table full, dropping packet”: new flows die. Timeouts: TCP established 432000 s (5 days), UDP 30 s, UDP stream 120 s. conntrack -L -E -S.
- A **flowtable** lets established forwarded flows skip the hooks from ingress.

Example — a namespace behind NAT (what runtimes do)

```
ip netns add blue # /var/run/netns/blue: only lo
ip link add veth0 type veth peer name veth1 # 1 cable, 2 ends
ip link set veth1 netns blue
ip addr add 198.51.100.1/24 dev veth0; ip link set veth0 up
ip -n blue addr add 198.51.100.2/24 dev veth1
ip -n blue link set veth1 up
ip -n blue route add default via 198.51.100.1
sysctl -w net.ipv4.ip_forward=1 # else nothing is forwarded
table ip nat { chain post { # loaded by nft -f
    type nat hook postrouting priority srcnat; policy accept;
    ip saddr 198.51.100.0/24 oifname "eth0" masquerade } }
```

Routing

- ip rule first: priority 0 local (table 255) · 32766 main (254) · 32767 default (253); selectors from to iif oif fwmark uidrange ipproto sport dport. Then longest prefix in the table; ip route get <dst> shows the kernel's pick.
- Types: unreachable (ICMP host unreachable), prohibit (admin. prohibited), blackhole (silent), throw (leave this table, try the next rule).
- rp_filter**: 0 off (kernel default), 1 strict, 2 loose (RFC 3704); the *max* of all and the interface applies; systemd ships 2. Strict + asymmetric routing = silent drops.

Layer 2 and namespaces

veth = a linked pair, the netns ↔ host cable; **bridge** = a software switch (bridge fdb); **VLAN**: ip link add link eth0 name eth0.10 type vlan id 10; **macvlan/ipvlan**: own address on the parent's LAN. A netns owns devices, routes, rules, sockets, /proc/net; when it dies, physical NICs return to the initial netns. br_netfilter loaded: bridge-nf-call-iptables (default 1) sends bridged IPv4 through iptables.

Sockets, buffers, queues

SO_RCVBUF/SO_SNDBUF: the request is capped at rmem_max/wmem_max (4 MiB in current docs), then **doubled** for bookkeeping — and that socket **stops autotuning**. TCP autotunes in tcp_rmem 4K / 128K / up to 32 MB by RAM (6 MB in the 6.12 docs) and tcp_wmem 4K / 16K / up to 4 MB. SO_REUSEPORT: many sockets, one port, kernel spreads the load. qdisc default pfifo_fast; multiqueue NICs get mq with it as leaves; systemd sets fq_codel. Congestion control default: **CUBIC**; bbr per sysctl or TCP_CONGESTION.

Tools

ip -br a -ip -s link	addresses · per-link drops (rtnetlink)
ss -tlnp -ss -tni	listeners + process · rtt, cwnd, retrans (sock_diag)
nstat -az TcpExtlisten*	ListenOverflows, ListenDrops
tcpdump -ni any port 443	any = every link; -w f.pcap to save
nft list ruleset conntrack -S	what is enforced · drop, early_drop

Traps

- ! “tcpdump saw it, so the firewall passed it” — RX taps run before tc, nft ingress and every IP hook; TX taps after the qdisc. 64 KB “packets” are GRO/TSO, not the wire.
- ! Docker -p is DNAT: traffic goes FORWARD, never INPUT, so ufw misses it — filter in DOCKER-USER. Docker that enables forwarding sets the FORWARD policy to DROP.
- ! tcp_tw_reuse does nothing for a server's TIME_WAITs; tcp_tw_recycle broke NAT'd clients and was removed in 4.12.